

Pranathi Kaparathi

+91 8520827413 kaparthipranathi@gmail.com LinkedIn GitHub

EDUCATION

Vellore Institute of Technology, Vellore (CGPA: 8.67)

B.Tech. - Computer Science(Internet Of Things)

2023 - 2027

Vellore, Tamil Nadu

Deeksha CFL PU College (Score: 93%)

Karnataka State Board (11th & 12th)

2021 - 2023

Bengaluru, Karnataka

Varin International Residential School (Score: 94%)

ICSE Board (10th)

2020 - 2021

Tumkur, Karnataka

SKILLS

Languages: Python, C/C++, SQL, JavaScript, TypeScript, HTML, CSS, Bash

AI/ML Frameworks: TensorFlow, Keras, PyTorch, Scikit-learn, XGBoost, HuggingFace Transformers

Data Science: NumPy, Pandas, Matplotlib, Seaborn, SciPy, LSTM Time-Series Forecasting, Feature Engineering

Deep Learning: CNNs, LSTM, Attention Mechanisms, Transfer Learning, Predictive Modeling, Anomaly Detection

IoT & Embedded: Arduino, ESP8266 (NodeMCU), Sensors (MQ135, MQ2, MQ7, DHT22), UART/I2C/SPI, Wi-Fi

Web & Backend: Node.js, Vite, REST APIs, Streamlit, Supabase (PostgreSQL), MongoDB, JWT Authentication

Tools & Platforms: Git, GitHub, Docker, Postman, Jupyter Notebook, Google Colab, VS Code, Linux

Concepts: DSA, OOP, DBMS, OS, Computer Networks, Edge AI, Fog Computing, Cloud Dashboard Integration

WORK EXPERIENCE

App & Web Development Intern | Spoorthy Integrated Solutions

Housekeeping Management Application

- Developed a digital housekeeping management application to streamline communication between housekeeping staff, supervisors, and clients.
- Enabled image-based reporting and monitoring of cleanliness status across facilities.
- Provided supervisors and clients with real-time visibility into housekeeping activities and task completion.
- Improved tracking, observation, and issue resolution through centralized reporting and workflow management.

PROJECTS

AI-Based Predictive Thermal & Load Management for Edge Computing | Python, TensorFlow, LSTM

- Designed an **LSTM-based time-series forecasting** system to proactively predict CPU/GPU temperature and load spikes across edge computing nodes before threshold violations occur.
- Built **simulation models** replicating real-world edge device behavior under varying workloads; generated synthetic datasets for model training and validation.
- Implemented **predictive load balancing algorithms** using NumPy, Pandas, and Scikit-learn, reducing simulated thermal violations by over **40%**.
- Evaluated performance with **MAE, RMSE, and R²** metrics; visualized real-time thermal trends via **Matplotlib** dashboards for interpretable operator insights.

AI-Powered Image Caption Generator | Python, TensorFlow, VGG16, MobileNetV2, LSTM, Streamlit

- Built end-to-end ML pipeline using **VGG16/MobileNetV2 CNNs** paired with **LSTM + attention mechanism** to generate human-like image captions with **95% accuracy**.
- Optimized inference speed by **40%** and reduced memory footprint by **60%** using lightweight MobileNetV2 feature extractors over heavier VGG16 architecture.
- Deployed as interactive **Streamlit web app** enabling real-time caption generation; improved accessibility for non-technical users with zero code interaction.

Intrusion Detection & Device Tampering System | Arduino, Embedded C, IoT Sensors

- Built embedded intrusion detection system with **PIR motion, reed switch, and vibration sensors**; programmed **interrupt-driven alert logic** in Embedded C triggering buzzer and LED alerts within milliseconds.
- Integrated **serial communication logging** for event timestamps and sensor state history enabling forensic-level audit trails; designed **low-power standby mode** for extended deployment lifetime.

Hybrid AI-Enabled Learning Platform | Vite, Node.js, Supabase, MongoDB, Gemini API

- Built full-stack platform integrating **Google Gemini API** for **NLP-based dynamic quiz generation** automatically creating contextual questions parsed from course content.
- Architected **hybrid database strategy** combining Supabase (PostgreSQL) for structured user/course data and MongoDB for flexible content storage, optimizing query performance across data types.
- Designed and exposed **RESTful APIs** for user authentication, course management, quiz delivery, and result tracking with **JWT-based** security for seamless frontend-backend communication.